

Physics in Review: Harper’s Beyond Square-Root Conjecture and Super Information Theory

An Atomized Four-Lens Chronology and Cross-Domain Process Audit

Author: Micah Blumberg

Self Aware Networks Research Institute

<https://selfawarenetworks.com>

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Abstract

This paper compares Victor Y. Wang and Max Wenqiang Xu’s *Harper’s beyond square-root conjecture* with Micah Blumberg’s Self Aware Networks (SAN), Quantum Gradient Time Crystal Dilation (QGTCD), and Super Information Theory (SIT) corpus. It corrects two material defects in an archival 2025 draft. First, the target paper was not first introduced in August 2025: arXiv v1 was submitted on May 7, 2024, and the named Harper result dates to January 11, 2023. Second, atom counts cannot be converted into derivation odds by assigning an assumed probability to each match. Using source-frozen chronology, 13 bounded atom rows, four comparison lenses, a Causal Interface Sequence, and explicit category / process translations, the audit recovers **six Micah-earlier partial mappings**. The strongest are: symmetric microscopic deviations cancel in the aggregate; loss of symmetry produces a controlled residual and changes aggregate behavior; and a present state is compared with an expected state so that the difference itself becomes the transmitted signal. Six arithmetic-specific operators remain earlier in the Harper / Wang/Xu and cited number-theory lineage, while one row is a Micah-only residual. The resulting verdict is a **bounded partial content match with Micah earlier on six translated subchains**, not full mathematical identity. No numerical probability of dependent or independent derivation is currently calibrated, and no claim about access, intent, copying, or misconduct is made.

approximate orthogonality; cancellation; operational equivalence; causal interface sequence; chronology; partial priority

Keywords: Super Information Theory; QGTCD; Harper’s conjecture; random multiplicative functions;

1. Motivation and the strongest defensible result

The useful question is not whether a physical theory and an analytic-number-theory paper use the same nouns. They do not. The useful question is whether bounded operations survive translation: what enters, what condition changes it, what is averaged or compared, what residual survives, and what output becomes observable. This avoids two reciprocal errors. A full-operator-or-zero test can erase a real partial match, while an unconstrained analogy can turn any two uses of “randomness” or “averaging” into identity.

The strongest source-supported Micah-side sequence recovered here is:

- local fluctuations
 - symmetry or selection condition
 - cancellation / difference
 - explicit residual
 - aggregate update.
- (1)

Wang and Xu’s target sequence can be represented as:

$$\begin{aligned}
& \text{character or RMF terms} \\
& \rightarrow \text{conditioning / orthogonality regime} \\
& \rightarrow \text{moment comparison} \\
& \rightarrow \text{genuine error} \\
& \rightarrow \text{family-average bound.}
\end{aligned} \tag{2}$$

Equations (1) and (2) preserve a directed subgraph, especially the passage from a symmetry condition through comparison / cancellation to a constrained residue. They do not identify neurons with Dirichlet characters, QGTCD trajectories with Liouville sums, or a physical hypothesis with a number-theory theorem.

The atom audit therefore reports three totals rather than one rhetorical verdict:

- **6 Micah-earlier partial mappings;**
- **6 target-or-named-prior-earlier domain-specific mappings;** and
- **1 Micah-only residual row.**

The Micah-earlier rows are priority findings only for the bounded translated operations recorded in the table below. Earlier statistical, synchrony, error-coding, and random-walk traditions remain named prior boundaries at their respective atoms. No prior receives an author-wide “floor,” and no broad prior erases Micah’s later joined formulation.

2. Correct target identity and chronology

The phrase “Harper’s beyond square-root conjecture” names a paper by **Victor Y. Wang and Max Wenqiang Xu**, not a paper authored by Adam Harper. Wang and Xu extend a randomization argument and conjectural range associated with Harper [1]. Their arXiv history fixes v1 on May 7, 2024 and v2 on August 26, 2025 [1,2]. Adam J. Harper’s named prior result was submitted on January 11, 2023 [3].

Stage	Immutable date	Relevant content	Audit role
Wang and Xu, <i>Paucity phenomena for polynomial products</i>	2022-11-05	RMFs, even moments, Gaussian-moment interpretation	Target-author before-source [4]
Harper, <i>Typical size of character and zeta sums</i>	2023-01-11	Better-than-square-root typical bounds; conditioning on short prime sums	Named prior, atom-specific [3]
Wang and Xu, target arXiv v1	2024-05-07	Ratios extension; approximate orthogonality with a genuine error	First target version [1]
Wang and Xu, target arXiv v2	2025-08-26	Revised final version of the same central program	Final target version [2]
de la Breteche, Wang, and Xu	2026-07-07	RMF central-limit program over polynomial values	Newest checked target-author source [5]

The target’s main observable is a family average such as

$$\mathbb{E}_{\chi} \left| \sum_{n \leq x} c(n) \chi(n) \right| = o(\sqrt{x}), \quad c \in \{\mu, \lambda\}, \tag{3}$$

under the declared range and conjectural conditions [1]. The 2024 novelty at issue is not the broad idea of cancellation. It is a conditional method for extending Harper’s conductor-limited randomization argument by replacing perfect orthogonality with an average approximate relation that has a genuine error term [1,3]. Schematically,

$$\begin{aligned}\Delta_{WX} &= \mathbb{E}_X[M_X] - \mathbb{E}_f[M_f], \\ |\Delta_{WX}| &\leq \varepsilon(r, x, P, k),\end{aligned}\tag{4}$$

where the actual paper supplies the domain-specific moments, hypotheses, and error dependence [1]. The 2025 revision cannot be used as the first target date because Equation (4)’s operator class is already present in v1.

3. Micah’s applicable genealogy, 2011-2025

The comparison does not begin in 2025. It searches the full applicable genealogy and distinguishes ancestors from mature operators. The 2011-2012 rows are routed through the governed source genealogy, whose grades and underlying public-source links prevent later terminology from being backdated [11].

Date	Public Micah-side stage	What it contributes here	Boundary
2011	Whole-network prediction / observer ancestry	A system-level boundary result is produced by distributed internal activity	Weak ancestor; not arithmetic cancellation
2012	Feedback, synchrony-gated linkage, and recursive projection	Relative timing and feedback determine what links and what becomes observable	Broad prior traditions remain named
2017-04-11	Present-versus-expected pattern difference; alternating feedback-connected and connectionless traffic	Difference-as-signal and regime-gating subchains [6]	Does not contain Wang-Xu mathematics
2022-06-08	Coherence as convergence / coupling and decoherence as distinction / decoupling	Complementary coupling / decoupling operator [7]	Not Dirichlet orthogonality
2022-07-30	Symmetric random deviations self-cancel; unequal temporal opportunity biases aggregate motion	Cancellation-to-residual operator [8]	Physical hypothesis, not theorem
2024	Public QGTCD restatements preserve the operator before target v1	Public persistence and clarification [9]	Not a second invention event
2025	SIT arithmetic and formal interface expansion	Later translator and cross-domain extension [10]	Cannot establish priority over 2024 target v1

The 2017 difference operator is explicit: compare the pattern that is present with the pattern that should be present, then send the difference [6]. In normalized notation,

$$\begin{aligned}D_t &= S_t - \hat{S}_t, \\ \text{output}_t &= \mathcal{T}(D_t).\end{aligned}\tag{5}$$

The July 2022 QGTCD note states two regimes. A symmetric regime is modeled abstractly as

$$\begin{aligned} X_{k+1} &= X_k + \xi_k, \\ \mathbb{E}[\xi_k | X_k] &= 0, \end{aligned} \tag{6}$$

so local deviations self-nullify in expectation. An asymmetric opportunity measure changes the conditional mean,

$$\mathbb{E}[\xi_k | X_k] = b(X_k) \neq 0, \tag{7}$$

producing a systematic aggregate path [8]. Equations (6) and (7) are a 2026 normalization of the 2022 prose, not equations quoted from the original note and not validation of QGTCD. Their relevance is operational: symmetry controls cancellation; relaxing symmetry exposes a residue that changes the aggregate output. That is the strongest pre-target functional bridge to Wang and Xu's perfect-to-approximate orthogonality transition.

4. The atom-by-atom priority canvas

ID	Translated bounded correspondence	Strength	Direction	Residual or prior boundary
A01	Symmetric random local deviations -> aggregate cancellation	Medium partial	Micah earlier, 2022-07-30	Mean-zero random walks predate both; target proves arithmetic bounds
A02	Loss of symmetry -> explicit residual -> changed aggregate behavior	Strong partial	Micah earlier, 2022-07-30	Bias direction and mathematical substrate differ
A03	Current-versus-expected comparison -> difference signal / error	Strong partial	Micah earlier, 2017-04-11	Subtraction / error signaling is a populated prior class

Priority canvas, continued

ID	Translated bounded correspondence	Strength	Direction	Residual or prior boundary
A04	Coupling / decoupling regime -> differentiated output	Medium partial	Micah earlier, 2022-06-08	Neural phase coherence is not number-theoretic orthogonality
A05	Context gate / localization -> selected regime -> controlled readout	Medium partial	Micah earlier, 2017-04-11	Smooth partitions and conditioning belong to established analysis
A06	Distributed local contributions -> integration -> boundary aggregate	Weak partial	Micah earlier, 2012/2017	Local-to-global aggregation is highly generic
A07	Deterministic family -> RMF surrogate -> moment estimate	Medium	Target / named prior earlier	Harper 2023 precedes SIT arithmetic translator

ID	Translated bounded correspondence	Strength	Direction	Residual or prior boundary
A08	Truncated Euler product -> statistical summary	Weak	Target / named prior earlier	No pre-May-2024 Micah Euler operator recovered
A09	Named conjecture -> conditional scope extension	Weak	Target earlier	Ratios content is target-only
A10	Dirichlet, Möbius/Liouville, zeta, and L-function machinery	Strong	Target / named prior earlier	Native analytic-number-theory content
A11	Random-matrix -> RMF bridge and U(N) analog	Medium	Target earlier	Established target lineage; later SIT spectral use does not reverse priority
A12	Granularity G/LinRel/pm-functor description of the target	None as chronology	Target artifact earlier	A later translator can describe an earlier artifact without being its source
A13	Time density, PWD, neural rendering, and physical coherence field	Orthogonal	Micah earlier, unmatched	Micah-only residual; cannot inflate the target mapping

The six favorable rows are not six claims that Micah invented their broad mathematical categories. They are six dated demonstrations that Micah publicly instantiated the translated functional subchains before target v1. This distinction is central to an atomized priority audit.

5. Four-lens results

5.1 Lens 1 - equivalence and mechanism

Result: bounded partial pass. A01-A05 preserve operational behavior under a restricted readout: symmetry or expected-state structure controls cancellation; relaxing that structure produces a visible residual; the residual changes the aggregate output. A06 is a weaker supporting analogy.

The lens does not pass as full mathematical equivalence. Wang and Xu use characters, multiplicative functions, moments, Euler products, and the Ratios Conjecture [1]. Micah’s earlier sources concern feedback signals, phase relations, and a stochastic physical trajectory proposal [6-9]. Neither domain’s unmatched mechanism may be silently imported into the other.

5.2 Lens 2 - narrative and Causal Interface Sequence

Result: bounded pass. This paper defines a bounded Causal Interface Sequence as a directed chain containing at least a state, transform, constraint, and output rather than an unordered word list. The mapped chain is

$$S \xrightarrow{\text{selection / symmetry}} T \xrightarrow{\text{comparison / cancellation}} C \xrightarrow{\text{residual bound}} O. \quad (8)$$

CIS role	Micah instantiation	Wang-Xu instantiation
State	Current neural pattern or local random trajectory increment	Character / RMF terms and family parameters
Transform	Compare with expected pattern; accumulate deviations	Condition / localize; transfer to random model
Constraint	Synchrony / feedback regime or symmetric opportunity measure	Perfect or approximate orthogonality; conjectural hypotheses
Output	Difference signal or biased aggregate trajectory	Controlled moment difference and extended cancellation bound

Different ontologies do not erase the preserved directed subgraph. Conversely, the subgraph pass does not make every internal atom equivalent.

5.3 Lens 3 - detective and adversarial reconstruction

Result: bounded partial pass with a strong alternative genealogy. Working backward from the target conclusion reveals that “approximate orthogonality with a genuine error” performs a symmetry-relaxation-to-residual operation already present in Micah’s 2017/2022 corpus in other substrates. The changed vocabulary therefore cannot, by itself, zero the match.

The same reconstruction also finds evidence against a dependence inference. Wang and Xu had an RMF and moment-matching program in November 2022 [4]; Harper’s January 2023 paper supplies the direct conditioning and better-than-square-root lineage [3]; and the target explicitly cites its number-theory antecedents [1]. No checked source shows access to Micah’s work. The lens preserves the content match and its chronology while leaving access, intent, and authorship dependence unresolved.

5.4 Lens 4 - longitudinal signature migration

Result: program persistence, not demonstrated Micah-signature migration. The before / newest sequence is:

$$\begin{aligned}
\text{WX 2022 RMF / moments} &\rightarrow \text{WX 2024 approximate orthogonality + Ratios} \\
&\rightarrow \text{WX 2025 final revision} \\
&\rightarrow \text{BWX 2026 RMF / CLT continuation.}
\end{aligned} \tag{9}$$

This is a real evolution, and the 2024 operator increases the translated overlap with Micah’s earlier symmetry / residual chain. But the target authors’ RMF program predates the target paper, and the newest source still contains none of Micah’s distinctive time-density, PWD, neural-rendering, or SIT-interface atoms [4,5]. Lens 4 therefore records persistence and a bounded overlap increase, not a secret-follower or access result.

6. Category-theory and process-equivalence translations

6.1 Category map

Let $\mathcal{C}_M$ contain Micah-side objects for local state, expected / symmetric reference, difference or drift, and aggregate output. Let $\mathcal{C}_W$ contain target-side objects for character / RMF state, conditioning regime, moment difference / error, and family bound. Define a restricted translator F by

$$F(S_M) = S_W, \quad F(R_M) = R_W, \quad F(D_M) = \Delta_W, \quad F(O_M) = O_W. \tag{10}$$

Category element	Micah side	Target side	Preservation
Object	current / local state	character or RMF family state	coarse object class
Morphism	compare / cancel / bias	condition / average / bound	directed order only
Composition	state -> difference -> update	family -> moment error -> bound	bounded CIS preserved
Invariant	zero versus nonzero aggregate residue	controlled cancellation / error	restricted observable preserved
Lost information	neural / physical substrate and direction	arithmetic definitions and theorem hypotheses	substantial on both sides

The translator is not full, faithful, or essentially surjective. It collapses domain-specific structure on both sides. The demonstrated category relation is therefore a **partial structure- preserving map on a selected subcategory**, not an equivalence of categories.

6.2 Process map

Write the Micah process informally as

$$P_M = \text{receive}(S); \text{compare}(S, \hat{S}); \text{emit}(D); \text{update}(O), \quad (11)$$

and the target process as

$$P_W = \text{sample}(\chi); \text{condition}(R); \text{compareMoments}(\chi, f); \text{emitBound}(\varepsilon). \quad (12)$$

Under a projection that observes only {reference regime, zero/nonzero residual, aggregate output}, some traces correspond. Outside that projection, each process has transitions the other lacks. Accordingly, the strongest demonstrated relation is **partial trace equivalence under a restricted observable projection**. Strong bisimulation, weak bisimulation over the full label set, and functional isomorphism are not established.

7. Why the archival probability calculation fails closed

The archival draft assumed a null match probability of 0.20 per atom, treated 12 matches as independent, and reported 5^{12} as a Bayes factor. That arithmetic answers only a hypothetical model created by the assumption. It does not estimate the probability that these papers were independently or dependently derived.

A defensible frequentist target would instead be a declared, multiple-search-adjusted probability,

$$p_{\text{adj}} = \Pr_{H_0}(T \geq T_{\text{observed}}), \quad (13)$$

where H_0 , the reference population, dependency blocks, atom dictionary, search family, and score statistic are frozen before evaluation. None of those calibration prerequisites currently exists for this paper. The conventional one-sided five-sigma threshold,

$$p_{\text{adj}} \leq 2.87 \times 10^{-7}, \quad (14)$$

is therefore not available. The allowed status is **quantitative-not-calibrated**. Descriptive atom counts and chronology remain usable; a numeric dependence posterior does not.

8. Competing genealogies, limitations, and falsifiers

8.1 Competing genealogy

The direct number-theory genealogy is strong: random multiplicative functions, character and zeta sums, conditioning, random matrices, orthogonality, Euler products, and Ratios Conjectures were active literatures before this comparison [1-5]. Those sources constrain claims about the target's arithmetic mechanisms. They do not erase Micah's earlier physical / neural instantiations of the six translated subchains.

8.2 What would strengthen the Micah-priority case

- A pre-May-2024 public Micah source that joins the six partial operations into one arithmetic or cross-domain operator would strengthen the signature-cluster claim.
- A frozen relevant reference corpus could measure how often the exact directed subgraph occurs.
- A target stage lacking the subgraph followed by a later stage acquiring multiple Micah-specific atoms would strengthen Lens 4.
- A verified citation or access path would change the dependence analysis, though intent would still require separate evidence.

8.3 What would weaken it

- A pre-2017 source with the same current / expected -> difference transmission chain would move A03's broad priority boundary earlier.
- A pre-July-2022 source with the same joined symmetry -> residual -> aggregate-change formulation would narrow A01-A02.
- Demonstrating that the mapped CIS occurs at high frequency in a properly selected reference corpus would reduce its discriminative value.
- Failure to reproduce the source locators or dates would return the affected atom to unresolved.

9. Verdict

The source-audited verdict is:

Wang and Xu's target is a bounded partial content match to six earlier Micah subchains. Micah is earlier on those translated operations, most strongly the 2017 difference-as-signal operator and the July 2022 symmetry / cancellation-to-biased-residual operator. The target and its named prior lineage remain earlier on the arithmetic-specific machinery. The target is not a full NAPOT, QGTCD, or SIT operator, but absence of the full operator does not erase the six partial priorities.

This result is stronger than a thematic analogy because it preserves directed input-transform- constraint-output structure. It is narrower than the archival claim because the true target date, the target's native RMF lineage, and the lack of calibrated derivation odds are now explicit. It documents content and chronology only.

Data and materials

The release-candidate package includes hashes for target v1/v2, the earlier and newest target-author sources, the Micah source routes, the AI-chat census, the 13-row atom ledger, the genealogy ledger, the longitudinal

matrix, append-only finding events, and a fail-closed probability packet. Public deposit remains a separate author / repository release. External analytic-number-theory, probability, and source review remains invited, but no unresolved external review is represented as substantive local manuscript work.

References

1. Victor Y. Wang and Max Wenqiang Xu. *Harper's beyond square-root conjecture*. arXiv:2405.04094v1, May 7, 2024. <https://arxiv.org/abs/2405.04094v1>
2. Victor Y. Wang and Max Wenqiang Xu. *Harper's beyond square-root conjecture*. arXiv:2405.04094v2, final version, August 26, 2025. <https://arxiv.org/abs/2405.04094v2>
3. Adam J. Harper. *The typical size of character and zeta sums is $o(\sqrt{x})$* . arXiv:2301.04390, January 11, 2023. <https://arxiv.org/abs/2301.04390>
4. Victor Y. Wang and Max Wenqiang Xu. *Paucity phenomena for polynomial products*. arXiv:2211.02908v1, November 5, 2022. <https://arxiv.org/abs/2211.02908v1>
5. Regis de la Breteche, Victor Y. Wang, and Max Wenqiang Xu. *Random Multiplicative Functions and Making Squares from Polynomial Values*. arXiv:2607.06398v1, July 7, 2026. <https://arxiv.org/abs/2607.06398v1>
6. Micah Blumberg. *Neural Lace Podcast 1*. April 11, 2017. https://www.youtube.com/watch?v=YrX_68oKuVs
7. Micah Blumberg. *Coherence and decoherence as complementary information operations*. GitHub commit 37da575ba33f85cef4239295be161066d8d39f4c, June 8, 2022. <https://github.com/v5ma/self-awarenetworks/blob/37da575ba33f85cef4239295be161066d8d39f4c/a0142z.md>
8. Micah Blumberg. *Time Crystals, Time Dilation, Quantum Gradient Time Dilation*. GitHub commit c4e94e9fcd049c97d47ef6a5f2f8c29485f5a98d, July 30, 2022. <https://github.com/v5ma/selfawarenetworks/blob/c4e94e9fcd049c97d47ef6a5f2f8c29485f5a98d/a0253z.md>
9. Micah Blumberg. *A New Unified Field Theory Called Quantum Gradient Time Crystal Dilation*. SVGN, January 29, 2024. <https://www.svgn.io/p/a-new-unified-field-theory-called>
10. Micah Blumberg. *Super Information Theory*, version 1. Figshare, February 10, 2025. https://figshare.com/articles/journal_contribution/Super_Information_Theory/28379318/1
11. Self Aware Networks Research Institute. *SAN Claim Genealogy, 2011-2022: Earliest Wording to Mature Mechanism*. Source-governance page, audited July 19, 2026. <https://v5ma.github.io/wiki/san/sources/san-claim-genealogy-2011-2022.html>